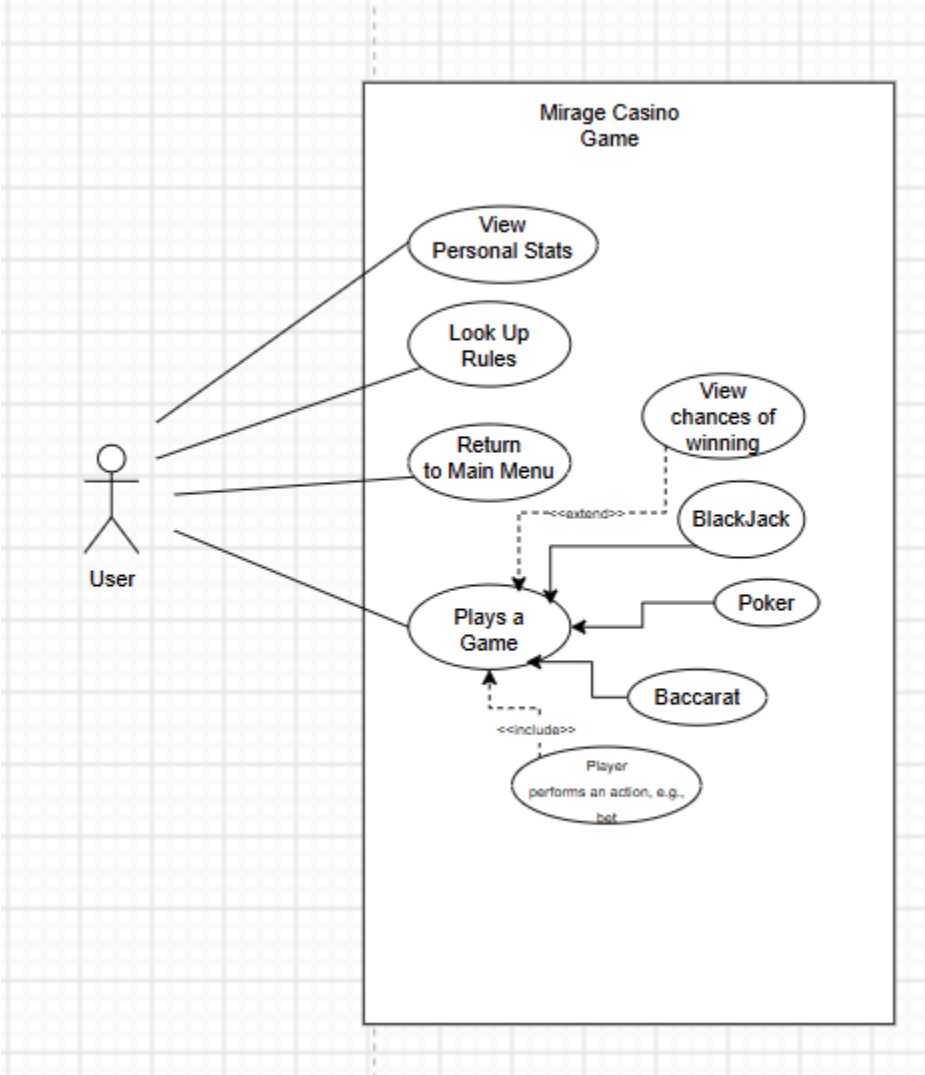
UML Class Diagram:

Link to UML Diagram: Untitled Diagram - Page-1

Use Case Diagram:



Use case #1: The user plays a full game of Blackjack.

- **Preconditions:** The casino game has to be running, the user has selected Blackjack from the main menu screen, and the Blackjack game has been created and the player is ready to perform an action.
- **Main Flow:** From the main menu, the user selects to play the Blackjack game. The game opens to the Blackjack page, with the main menu disappearing. 2 cards are laid out for

the Player and Dealer, with the Player being able to hit or stand. The Player hits, and another card is added to the Player's hand. If the Player's hand is less than 21, the Player is able to hit or stand. If the Player's total hand is above 21 the Player loses. If the Player chooses to stand the Dealer flips the card which is turned over, and from there can choose to hit or stand. The Dealer completes their chosen turn and the results are calculated. If the Player and Dealer both have not busted and they both stand their individual scores are computed, the winner being whoever has the higher score. The system displays the result, and the user returns to the main menu.

- **Alternative scenarios**: The player busts, causing the Dealer to win. Another alternative scenario is if the Player lands a perfect Blackjack (an ace paired with any face card) which causes them to automatically win (unless the Dealer also has Blackjack).
- **Postconditions:** The Blackjack round has ended, the Player's and Dealer's scores are calculated, the game determines an outcome and the result is displayed back to the user.
- **Exception Flow:** The Blackjack game does not show the option to play again. In this scenario the game would abruptly stop and return to the main menu.

Use case #2: The user plays a full game of Poker.

- **Preconditions:** The casino game has to be running, the user has selected Poker from the main menu screen, and the Poker game has been created and the player is ready to perform an action.
- **Main Flow:** From the main menu, the user selects to play the Poker game. The game opens to the Poker window, with the main menu disappearing. 2 cards are laid out for the Player and Dealer with the Player being able to bet or fold. The Player chooses to bet and so does the Dealer, and from there 3 cards are placed in the flop. It is now the Player's turn so the Player chooses to check, and so does the Dealer. The 4th card is now dealt and revealed onto the flop, and the Player and Dealer both choose to check again. The same process is repeated for the 5th and final card. Once both the Dealer and Player have checked, they both reveal their hands. Based on the rankings of hands in Poker, either the Player or Dealer is determined to be the winner. The system displays the result, and the user returns to the main menu or to play again.
- **Alternative scenarios:** The player or dealer bet instead of checking, causing the pot to increase. Another example is that the Player can fold their cards if they don't like the odds instead of betting or checking.
- **Postconditions:** The Poker round ends, and the winner is determined. The pot is then given to the winner, with the chip balance of both the Player and Dealer being updated according to how much money they bet. The results are displayed, and the user is prompted with the option to either Play Again or Return to Main Menu.
- **Exception Flows:** The cards are not dealt in the community card section when the bet is placed from the Player. This will result in the system displaying an error message after waiting a certain amount of time, and returning to the main menu.

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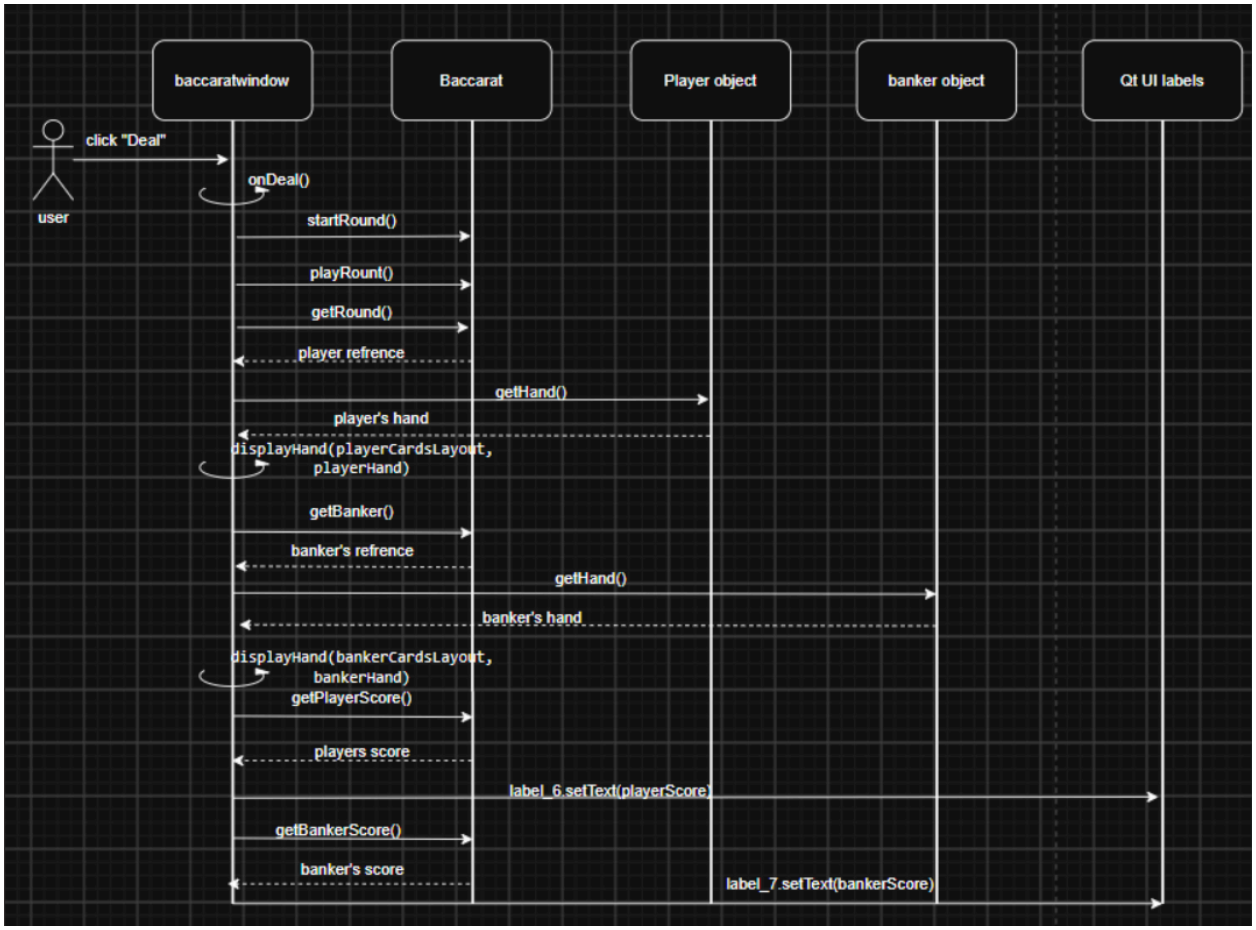
User Case #3: Restarting a Game

- **Preconditions:** The game must be initialized and playable when the user presses that game's button.
- **Description:** The player can't restart a game until the game is finished. When the Player clicks restart the system resets the current game state back to the default with an empty player hand and all attributes (eg chips, points) back to the loaded state. The game also resets the interface as well so the player can start a fresh game. The conditions are that a Game (Poker, Blackjack, Baccarat) was started, running and finished. Then the Player has to click the "Play Again" button to restart the game. Internally, the system restarts the game by using GameManager, so when the "Play Again" button is clicked the onPlayAgain method is called which calls GameManager to restart the round.
- **Alternate Flows:** Instead of the user clicking the "Play Again" button when the game ends, the user clicks the "Main Menu" button which brings them back to the main menu.
- **Exception Flows:** The round the user is playing does not end and continues indefinitely without a winner being determined and a game being reset.

Sequence Diagram:

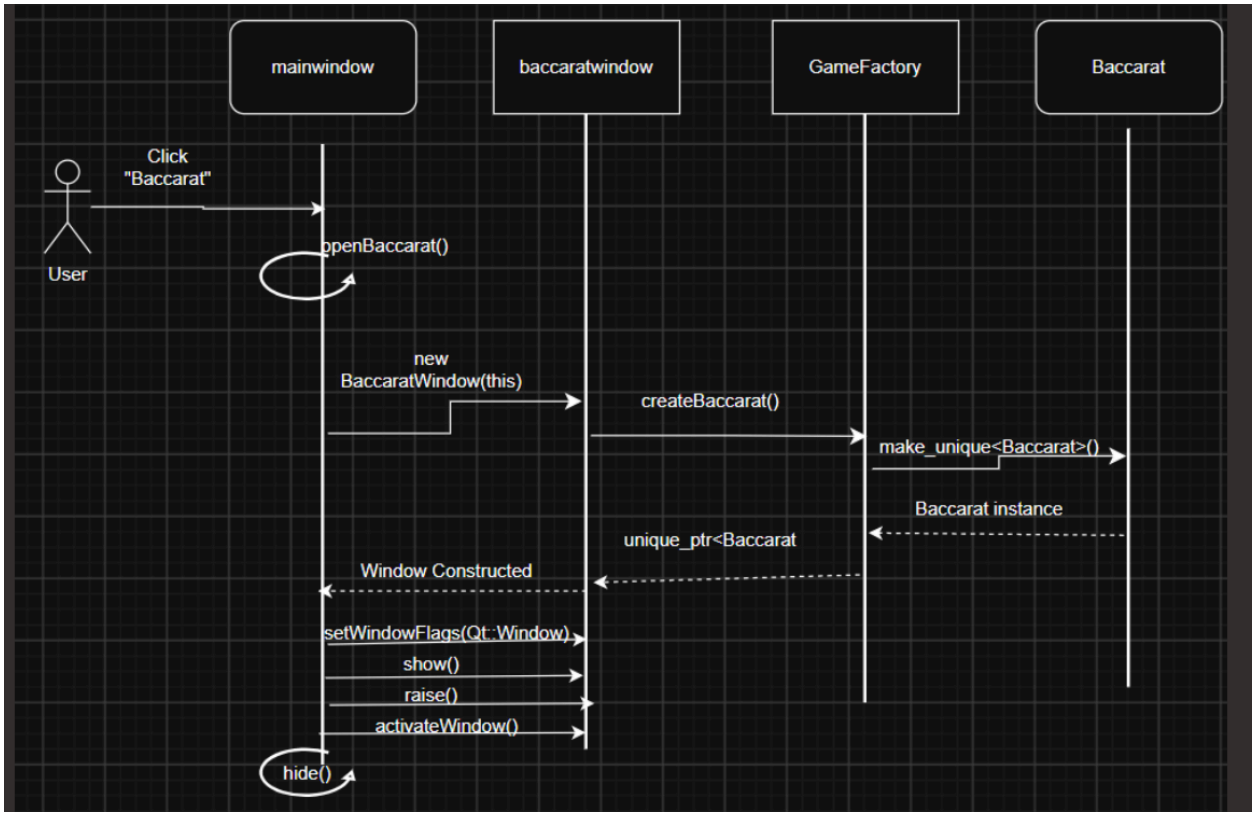
Sequence Diagram #1:

-



Sequence Diagram #2:

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mainwindow
baccaratwindow
GameFactory
Baccarat
Click
"Baccarat"
User
openBaccarat()
new
BaccaratWindow(this)
createBaccarat()
make_unique<Baccarat>()
Baccarat instance
unique_ptr<Baccarat
Window Constructed
setWindowFlags(Qt::Window)
show()
raise()
activateWindow()
hide()